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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

BT1301-1 – BIOFUELS ENGINEERING

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The Bioenergy is of biological origin by recent carbon
 - A) Fixation
 - B) Degradation
 - C) Reduction
 - D) Oxidation
2. First generation biofuels pose major threat to
 - A) Fuel supply
 - B) Chemical supply
 - C) Medicine supply
 - D) Food supply
3. Hemi cellulose is ____% of the total dry weight of woody biomass
 - A) 40-50
 - B) 25-35
 - C) 50-60
 - D) 60-70
4. This biofuel is proposed to produce from extracting oil of algae is
 - A) Third generation
 - B) First generation
 - C) Second generation
 - D) Fourth generation
5. The vegetable oil _____ times larger than the normal biodiesel molecule.
 - A) 2
 - B) 3
 - C) 4
 - D) 5
6. If the trace of water is left in the vegetable oil transesterification then _____ is formed.
 - A) Ester
 - B) Soap
 - C) Biodiesel
 - D) Diesel
7. Bio ethanol is denatured alcohol that is also called as
 - A) Methylated spirit
 - B) Ethylene glycol
 - C) Ethylene
 - D) Methylene
8. To make transport fuel the bio ethanol is blended with
 - A) Kerosene
 - B) Petrol
 - C) Vegetable oil
 - D) Natural gas
9. Water will be the coal of the future the statement is by
 - A) James Watt
 - B) Newton
 - C) Einstein
 - D) Jules Vernes
10. In hydrogen producing microorganism eukaryotes include
 - A) Cyanobacteria
 - B) Bacteria
 - C) Algae
 - D) Virus
11. Psychrophilic bacteria temperature range (degree Celsius)
 - A) minus 10 to +20
 - B) minus 10 to +30
 - C) minus 10 to +40
 - D) minus 10 to +10
12. Biological hydrogen production reactions involve these enzymes
 - A) Hydrogenase or Nitrogenase
 - B) Reductase or Hydrogenase
 - C) Oxidase or Hydrogenase
 - D) Reductase or oxidase

13. In Anode chamber of MFC electron donor compound is
 - A) Evaporated
 - B) Reduced
 - C) Condensed
 - D) Oxidized
14. Under normal aerobic growth conditions _____ microbes oxidize chemical compounds
 - A) Chemotrophic
 - B) Lithotrophic
 - C) Mesophilic
 - D) Thermophilic
15. In single chamber designs, wet-proofed which cloth is used in cathode
 - A) Nitrogen
 - B) Carbon
 - C) Nylon
 - D) Polyester
16. Protons produced from the oxidation of substrate in the anode compartment
 - A) Decreases the pH
 - B) Increases the pH
 - C) Decrease and then increases pH
 - D) Stable pH
17. Which types of Biogas plant is fed and emptied regularly?
 - A) Continuous type plants
 - B) Batch type plants
 - C) Dome type plants
 - D) Drum type plants
18. Which is the third stage of anaerobic digestion?
 - A) Acidogenesis
 - B) Methanogenesis
 - C) Acetogenesis
 - D) Hydrolysis
19. The process of burning of municipal solid waste at high temperature is called
 - A) Composting
 - B) Incineration
 - C) Land filling
 - D) Shredding
20. Which of the following process takes place in closed container in absence of oxygen?
 - A) Pyrolysis
 - B) Combustion
 - C) Liquefaction
 - D) Gasification

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Comparatively explain the role of biofuels in mitigating detrimental effects of fossil fuels.	4	L*1	1	1
	b) Explain how energy is produced by utilizing agro-industrial waste.	4	L2	1	1
	c) Describe the transesterification process of biodiesel production with flowchart.	8	L2	2	1,2
2.	a) Write a brief account on any two starch crops for biofuel production.	4	L1	1	1
	b) Explain sewage as a source for biofuel production.	4	L2	1	1
	c) Describe any four physico-chemical characterization tests for biodiesel feedstock oil.	8	L2	2	1,2
3.	a) With reference to the raw materials for biodiesel production, discuss any two non edible seed oil sources and waste cooking oil.	8	L2	1	1
	b) Describe the fermentation process of bioethanol production with flow chart.	8	L2	2	2
Unit – II					
4.	a) Elaborate on raw materials used for bio-hydrogen production.	8	L1	3	1
	b) With illustration explain single chamber and dual chamber microbial fuel cell (MFC) designs.	8	L2	4	2
5.	a) Explain the application of any two reactor configurations for hydrogen production.	8	L2	3	2
	b) Explain the application of MFC in waste water treatment and its benefits.	8	L2	4	2,5

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| 6. | a) | Describe the factors that play a major role in hydrogen production. | 8 | L2 | 3 | 2 |
| | b) | Write briefly about the advancement in microbial fuel cell (MFC). | 8 | L3 | 4 | 1 |

Unit – III

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|----|----|--|---|----|---|-----|
| 7. | a) | Explain two main types of biogas digester models with neat labelled diagram. | 8 | L2 | 5 | 1,6 |
| | b) | Describe Fluidized bed incineration systems. | 8 | L2 | 5 | 2 |
| 8. | a) | Explain the thermochemical pyrolysis process for biomass. | 8 | L2 | 5 | 2 |
| | b) | Describe the hydrothermal processing technology (HTL) for biomass feedstock. | 8 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
